

FR-017 M2 — Corpus Reconstructability & Deterministic Evaluation

FR017 — sources/optional/m2.md

Masterclass source package snapshot

Functional Requirement	FR-017
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete (M2)** — **GO** (derive-only; reconstructability reliable)
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via python scripts/build_masterclass_package.py. Repository documentation remains the source of truth.

Contents

FR-017 M2 — Corpus Reconstructability & Deterministic Evaluation

Status: **Complete (M2)** — **GO** (derive-only; reconstructability reliable) (**Accepted** — unlocked M3);
FR-017 later **Complete / Frozen** — [fr017_agent_evaluation_observability.md](#)

Date: 2026-08-07

Phase: Horizon 1A Stage 11

Architecture: [ADR-009](#)

Preceding: [M1 contracts](#) (**Accepted**); [M0 spike](#) (**Accepted**)

Succeeded by: [M3 owner CLI](#) (**Complete**)

Does not begin: M4 freeze, dashboards, telemetry stores, tracing platforms, DOS / BOPA / OBS / orchestration runtime changes, Horizon 1B.

1. Executive summary

M2 packages a **deterministic, derive-only** observability corpus over static `OrchestrationRun` / `Handoff` / optional `AgentRunMetrics` fixtures. It exercises M1 metrics, parent/child correlation, orphan detection, missing-vs-zero semantics, R1–R12 reconstructability, and mixed-corpus aggregation — **without** invoking DOS, BOPA, OBS, or adding events/SoTs.

Signal	Result
Cases	15/15 passed
Deterministic repeat	Yes
Runtime instrumentation required	No
FR-016 redesign required	No
Horizon 1B blocked	No
Go / no-go	GO

Useful evaluation remains derive-only. Reconstructability is reliable from existing records. **Do not DEFER FR-017** on instrumentation grounds.

2. Corpus design

Module: `career_intelligence.multi_agent.observability_corpus`

API: `run_observability_corpus()` → `ObservabilityCorpusReport`

Fixtures use **fixed ULID ids and timestamps** so dumps are byte-stable across runs. No supervisor execution; no live LLM; no store I/O.

Case	Intent
C01	Complete successful OBS run
C02	Delegation blocked (deny + <code>delegation_blocked</code>)
C03	BOPA child run with provider/token/cost
C04	OBS brief (<code>brief</code> goal)
C05	<code>prepare_then_brief</code> — BOPA + OBS outputs
C06	Missing optional provider/model/token/cost/latency (<code>None</code>)
C07	Measured zero distinguished from missing
C08	Orphaned child reference (parent $\neq$ handoff)
C09	Missing child <code>AgentRunMetrics</code> join (ids present)
C10	Stale / incomplete handoff (<code>handoff_stale</code>)
C11	Safe resume — hash change + idempotency; single visit
C12	Loop stops: <code>repeated_delegation</code> , <code>circular_delegation</code> , <code>no_progress</code>
C13	Provider unavailable
C14	Malformed / contradictory audit linkage
C15	Mixed-corpus aggregation

3. Metrics derived

Per case (via `extract_orchestration_run_metrics` / `HandoffMetrics`):

- identity / goal label / status / stop reason / owner action
- step and visit utilisation + limit flags
- handoff counts (allow/deny) and per-handoff policy/lifecycle
- observation snapshot fields (hash, package/truth/pipeline/readiness)
- child ids + optional FR-015 `AgentRunMetrics` roll-up
- optional provider / model / tokens / cost / elapsed (`None` when absent)

C15 aggregate example (acceptance run): 8 fixture runs mixed into one corpus — handoff totals, deny counts, `provider_unavailable_count=1` , token totals only where present.

4. R1–R12 results

Case	R failures (expected)	Notes
C01–C07, C09–C13	none	Full reconstructability
C08	R11	Orphan parent/handoff child ids
C12 (×3 fixtures)	none	Stop reasons reconstructable
C14	R7, R11	Completed OBS without brief ref + orphan parent child / brief
C15	n/a (aggregate)	Per-member R checked in source fixtures

Happy-path and deny/stop paths answer R1–R12 from audits alone. Gap cases **surface** R failures rather than inventing evidence — correct derive-only behaviour.

5. Correlation / orphan results

Case	correlation_complete	Orphans
C01–C07, C09–C13	true	none
C08	false	parent <code>AGR_ORPHAN</code> ; handoff <code>AGR_1</code>
C14	false	parent <code>AGR_1</code> ; brief <code>OBR_1</code> not on handoff
C12 fixtures	true	deny handoffs without children

6. Missing-versus-zero validation

Case	Result
C06	<code>provider / model / tokens / cost</code> all None
C07	<code>tokens / cost</code> 0 / 0.0 ; distinct from C06
C09	join absent → tokens None (ids still correlate)
C15	missing-only sub-aggregate keeps <code>total_input_tokens</code> is None

Counts remain **0** for empty handoff/event sets. No coercion of missing → zero.

7. Deterministic repeatability

`run_observability_corpus` runs the suite twice and compares:

- per-case `passed` / `case_id`
- JSON dumps of all `OrchestrationRunMetrics`
- JSON dumps of all `ReconstructabilityReport`s

Acceptance run: `deterministic_repeat_ok=True`.

8. Defects or gaps

Item	Disposition
Illegal Pydantic shapes rejected at FR-016 write time	Out of scope — write validators remain FR-016; C14 uses valid but contradictory linkage
Early <code>running</code> with no selection may fail R3	Documented M1 limitation; not required for M2 corpus terminals
Live FR-016 A–T supervisor corpus not re-executed	Intentional — M2 is audit-fixtured eval; FR-016 <code>run_corpus</code> remains unchanged
Metrics CLI	Deferred to optional M3
No defect requiring DOS/BOPA/OBS change	Confirmed

9. Product and learning value

Lens	Assessment
Commercial / job-search effort	Low direct value (eval substrate)
Learning / interview transfer	High — derive-only observability, reconstructability as acceptance, missing≠zero
Daily prep path	Unchanged — prefer <code>cic agent run</code>
Horizon 1B	Unblocked

10. M2 go / no-go

Verdict: GO

Gate	Result
Useful evaluation remains derive-only	Yes
Reconstructability reliable from existing records	Yes
Requires runtime instrumentation / new events	No → would have been DEFER
Requires dashboards / telemetry store	No
Requires FR-016 redesign	No

DEFER criteria not met. Proceed to optional M3 only on owner request; otherwise FR-017 may pause while Horizon 1B proceeds.

11. Tests

Suite	Path	Result
M1 contracts	tests/unit/multi_agent/test_observability_m1.py	pass
M2 corpus	tests/unit/multi_agent/test_observability_corpus_m2.py	pass

Harness: `run_observability_corpus / go_no_go_observability`.

12. Final repository status

Item	Status
FR-017 M0	Accepted
FR-017 M1	Accepted / frozen contracts
FR-017 M2	Complete — GO
FR-017 M3	Not started (owner request)
FR-017 M4	Not started
Horizon 1B	Not blocked
DOS / BOPA / OBS / runtime	Unchanged
ADR-009	Remains binding

Owner next step: Accept M2 GO unlocked **M3** ([fr017_m3_owner_cli.md](#)). Accept M3 to unlock optional M4 — or defer remaining FR-017 and begin Horizon 1B.

Do not implement M4 without explicit owner request.